M4: Short Answers on Rapidity

A Teaching Walkthrough

PHYS 7502 — Prof. Frantz

Before We Start: The Three Formulas

The entire problem set is about comparing three closely related formulas. Let's lay them out clearly, because if you understand how they relate, every answer falls out naturally.

Formula 1: Our Class Definition of Rapidity y

$$\boxed{y \;=\; \frac{1}{2}\ln\!\left(\frac{E+p_z}{E-p_z}\right) \;=\; \operatorname{arctanh}\!\left(\frac{p_z}{E}\right) \;=\; \operatorname{arctanh}(\beta\cos\theta)}$$

This uses p_z, the momentum component **along the beam**. It tells you how much of this particle's motion is *along the beam direction*. A particle at rest has $y = 0$. A particle flying along the beam has $|y| \gg 1$. A particle flying sideways ($\theta = 90°$) also has $y = 0$.

Formula 2: Pseudorapidity η

$$\boxed{\eta \;=\; -\ln\!\left[\tan\!\left(\frac{\theta}{2}\right)\right] \;=\; \operatorname{arctanh}\!\left(\frac{p_z}{|\vec{p}\,|}\right)}$$

This is just a remapping of the polar angle θ. It needs **no mass, no energy**—just the direction the particle went. That's why experimentalists love it: you only need a tracker, not a calorimeter.

Formula 3: Textbook Rapidity (Mann, L&B)

$$\boxed{\gamma = \cosh y, \qquad \beta = \tanh y, \qquad \beta\gamma = \sinh y}$$

This parametrizes the **total speed** β of a 1D Lorentz boost, or of a particle whose momentum is entirely in one dimension.

Intuition

Now stare at Formulas 1 and 2 side by side:

$$y = \operatorname{arctanh}\left(\frac{p_z}{\boldsymbol{E}}\right) \qquad \text{vs.} \qquad \eta = \operatorname{arctanh}\left(\frac{p_z}{|\boldsymbol{\vec{p}}|}\right)$$

Same function. Same numerator. The *only* difference is what sits in the denominator: E or $|\vec{p}|$. Since $E = \sqrt{|\vec{p}|^2 + m^2}$, we have $E \geq |\vec{p}|$, with equality when $m = 0$.

That one substitution—E vs. $|\vec{p}|$—is the key to parts (a) and (b).

Part (a): How Our Definition Differs from the Textbook

What is this question really asking?

Two different formulas both go by the name "rapidity." How are they different?

Our version: $\tanh y = \beta \cos\theta = \beta_z$ (*longitudinal* velocity only)

Textbook version: $\tanh y = \beta$ (*full* speed)

Under what condition are they the same?

Think of it this way. In the textbook treatment, rapidity is a convenient way to label *how fast something is going*. Instead of β, which has a maximum of 1, you use $y = \text{arctanh}(\beta)$, which runs from 0 to ∞. It's the same information, just a nicer parameterization—$\gamma = \cosh y$ is easier to manipulate than $\gamma = 1/\sqrt{1-\beta^2}$.

But in a real experiment, the particle isn't moving in one dimension. It has a 3D momentum vector at some angle θ to the beam. Our class definition says: *I only care about the beam-direction component.* So instead of the full speed β, we plug in $\beta_z = \beta\cos\theta$—the fraction of the velocity that's along the beam.

The consequence is immediate:

$$\tanh y_{\text{class}} = \beta\cos\theta \qquad \neq \qquad \tanh y_{\text{textbook}} = \beta$$

They agree **only when** $\cos\theta = 1$, i.e., the particle flies exactly along the beam ($\theta = 0°$ or $180°$, meaning $p_T = 0$).

There's a clean algebraic way to see the same thing. Our definition gives:

$$E = m_T \cosh y, \qquad p_z = m_T \sinh y \qquad \text{where } m_T = \sqrt{m^2 + p_T^2}$$

The textbook gives:

$$E = m\cosh y, \qquad |\vec{p}| = m\sinh y$$

These are the same only if $m_T = m$ and $p_z = |\vec{p}|$, which both require $p_T = 0$.

Answer

The class definition uses only the *longitudinal* velocity component ($\beta_z = \beta\cos\theta$), so it works for a particle flying at any angle relative to the beam. The textbook definition parameterizes the *total* speed (β), treating rapidity as a property of a 1D boost. They give the same number only when the particle moves exactly along the beam axis ($\theta = 0°$ or $180°$, i.e. $p_T = 0$).

Part (b): Are η and y Numerically the Same?

What is this question really asking?

Sub-question 1: If you compute both η and y for the same particle at a collider, how close are the two numbers? Choose from: very different / different / similar / effectively the same / almost exactly the same / exactly the same.
Sub-question 2: Which variable is better for defining a "maximum interesting value"—y or η?

Sub-question 1

We already identified the key: y has E in the denominator, η has $|\vec{p}|$. So the question boils down to: *how different are E and $|\vec{p}|$ for a typical particle at a collider?*

$$E = \sqrt{|\vec{p}|^2 + m^2} \approx |\vec{p}| + \frac{m^2}{2|\vec{p}|} + \cdots \quad \text{when } |\vec{p}| \gg m$$

At the LHC, a "typical" pion ($m \approx 140\,\text{MeV}$) might have $p \sim 5\,\text{GeV}$. So $E/|\vec{p}| \approx 1 + (0.14)^2/(2 \times 5^2) \approx 1.0002$. The difference is two hundredths of a percent! For a proton at the same momentum, $E/|\vec{p}| \approx 1.009$, still less than 1%.

The upshot: for the particles you actually detect at a modern collider, E and $|\vec{p}|$ are essentially identical, so y and η are essentially identical. The answer is **effectively the same**.

There are two footnotes to be honest about. First, for **massless** particles (photons, neutrinos), $E = |\vec{p}|$ exactly, so $y = \eta$ *exactly*. Second, the approximation breaks down at very forward/backward angles (large $|y|$) for massive particles—that's where the small $E/|\vec{p}|$ difference gets amplified by the arctanh. But in the central detector region, the difference is negligible for any particle species.

Sub-question 2

Pseudorapidity for the beam particle itself is $\eta_{\text{beam}} = -\ln[\tan(0/2)] = \infty$. That's not a useful maximum.

Rapidity of the beam is $y_{\text{beam}} = \text{arctanh}(\beta_{\text{beam}})$—a finite, meaningful number. At RHIC (100+100 GeV gold beams), $y_{\text{beam}} \approx 5.4$. At the LHC (6.5+6.5 TeV protons), $y_{\text{beam}} \approx 9.5$.

Produced particles generally have $|y| \leq y_{\text{beam}}$ (by conservation laws—you can't get more longitudinal momentum out than what went in). So y_{beam} sets a natural, physical boundary on the

interesting kinematic range.

Therefore: **rapidity y is preferred** when considering maximum values.

Answer

(b.1): η and y are **effectively the same** for practical purposes. The only difference (E vs. $|\vec{p}|$) is negligible at collider energies where $|\vec{p}| \gg m$ for essentially all detected particles. They are exactly equal for massless particles.

(b.2): Rapidity y is preferred for maximum values. The beam rapidity $y_b = \text{arctanh}(\beta_{\text{beam}})$ is finite and physically meaningful (it bounds the kinematic range of produced particles), while $\eta_b \to \infty$ since $\theta_{\text{beam}} = 0$.

Part (c): Two Advantages Over θ

What is this question really asking?

Give two concrete reasons why collider physicists use y or η instead of θ.
For each, state whether it applies to y only, η only, or both.

Advantage 1: Rapidity differences are Lorentz invariant

This is the single most important property. Under a Lorentz boost along the beam axis with velocity β_F, rapidity transforms by **simple addition**:

$$y' = y + y_F, \qquad \text{where} \quad y_F = \operatorname{arctanh}(\beta_F)$$

This was derived in class by applying the Lorentz transformation to E and p_z, inserting into the definition of y', and watching the γ factors cancel (they factor as a product inside the log and split into a sum).

The constant y_F is the same for every particle—it depends only on the boost. So if you take the rapidity difference between any two particles:

$$\Delta y \;=\; y_2 - y_1 \;=\; (y_2' - y_F) - (y_1' - y_F) \;=\; y_2' - y_1' \;=\; \Delta y'$$

It's invariant. This has huge consequences:

- Distributions dN/dy keep the **exact same shape** in any boosted frame. The boost just slides the histogram left or right.
- Jet cone sizes ($\Delta R = \sqrt{\Delta y^2 + \Delta\phi^2}$) don't change under boosts.
- You can directly compare results between fixed-target experiments (large boost) and colliders (small or zero boost).

The angle θ has none of this. Under a boost, θ changes via relativistic aberration—a messy, energy-dependent transformation that is different for every particle.

Applies to: Strictly y **only**. Pseudorapidity η is a geometric quantity and does not transform additively under boosts. However, since $y \approx \eta$ in practice, $\Delta\eta \approx \Delta y$ is *approximately* invariant—good enough for most real analyses.

Advantage 2: Particle distributions are approximately flat

At a high-energy collider, if you histogram all produced particles vs. θ, you see a catastrophic spike near 0° and 180°—essentially all the particles crowd into a tiny angular region near the beam. The plot is useless.

Histogram the same particles in y or η, and the distribution "unfolds" into a broad, gentle, roughly **flat plateau** in the central region. This is a direct consequence of Advantage 1: particle production is approximately isotropic in the center-of-mass frame, and the rapidity variable preserves that uniformity rather than distorting it.

The flat shape means equal-width bins in y or η contain comparable numbers of particles—a basic requirement for sensible data analysis.

Applies to: Both y and η, since $dN/d\eta \approx dN/dy$ at high energies.

Answer

Advantage 1: Rapidity differences Δy are Lorentz invariant under longitudinal boosts ($y' = y + y_F$), so particle distributions keep the same shape in any frame and angular sizes like jet cones are frame-independent. Strictly applies to y only; approximately to η.

Advantage 2: Particle distributions dN/dy and $dN/d\eta$ are roughly flat over a broad central region, making them practical for binning and displaying data. In contrast, $dN/d\theta$ is catastrophically peaked at forward/backward angles. Applies to both y and η.

Part (d): Why the "Units" Are Radians

What is this question really asking?

People say "one unit of rapidity ≈ one radian."

(d.1) Where does this come from?

(d.2) Is it exactly true everywhere, or only at a special angle?

(d.3) What does $\Delta y = 1$ between two particles (or two frames) physically mean?

Step 1: The Jacobian at 90°

Start with $\eta = -\ln[\tan(\theta/2)]$ and take the derivative:

$$\frac{d\eta}{d\theta} = -\frac{1}{\sin\theta}$$

(Chain rule: the derivative of $-\ln[\tan(\theta/2)]$ is $-\frac{1}{\tan(\theta/2)} \cdot \frac{1}{2\cos^2(\theta/2)} = -\frac{1}{2\sin(\theta/2)\cos(\theta/2)} = -\frac{1}{\sin\theta}$.)

Now evaluate at $\theta = 90°$ (midrapidity, $\eta = y = 0$):

$$\left|\frac{d\eta}{d\theta}\right|_{\theta=90°} = \frac{1}{\sin 90°} = 1$$

This means: at $\theta = 90°$, a small change $d\eta$ equals the same small change $d\theta$ in radians. **One unit of η = one radian**, exactly, at midrapidity.

Step 2: Away from 90°, it breaks in the lab—but not in the COM

At other angles, $|d\eta/d\theta| = 1/\sin\theta \neq 1$. At $\theta = 5°$, for instance, $|d\eta/d\theta| \approx 11.5$, so one unit of η spans only $\sim 5°$ of lab angle. At $\theta = 0.1°$, one unit of η is milliradians. In the lab frame, the "1 unit = 1 radian" rule clearly fails away from 90°.

But now invoke Advantage 1: Δy is Lorentz invariant. A particle pair at high rapidity (near the beam, tiny lab angle) can be boosted to a frame where they're at $y \approx 0$. In *that* frame—effectively the center-of-mass frame of whatever process produced them—they're near $\theta = 90°$, and $\Delta y = 1$ corresponds to $\Delta\theta = 1$ radian. Since Δy doesn't change under the boost, the *COM-frame* interpretation holds everywhere:

Key Takeaway

$\Delta y = 1$ between two particles always means they are separated by approximately **one radian** ($\approx 57°$) **in the center-of-mass frame**, regardless of their lab-frame angle.

Step 3: Concrete examples

This gives rapidity a visceral, angular meaning:

- A jet with cone size $\Delta R = 0.4$ subtends about $0.4\,\text{radians}$ ($\sim 23°$) of COM-frame angle.
- Two back-to-back particles from a decay have $\Delta y \sim \pi$ ($\sim 180°$) in the COM.
- The LHC at $\sqrt{s} = 13\,\text{TeV}$ has $y_{\text{beam}} \approx 9.5$. That means the forward beam direction has been "stretched" to $\sim 9.5\,\text{radians}$ ($\sim 3\pi$) of effective angular space along the beam axis—the Lorentz boost is packing huge amounts of COM-frame angular range into a tiny lab-frame cone.

Answer

At $\theta = 90°$, differentiating the pseudorapidity definition gives $|d\eta| = |d\theta|$, so one unit of η *exactly* equals one radian there. Away from 90° the lab-frame correspondence breaks down ($|d\eta/d\theta| = 1/\sin\theta \neq 1$), but the Lorentz invariance of Δy rescues the interpretation: since any high-rapidity particle pair can be boosted to the COM frame where they sit at $\theta \approx 90°$, the meaning "$\Delta y = 1 \approx 1\,\text{radian}$ in the COM" holds universally.
A rapidity difference Δy between two particles, two directions, or two reference frames always corresponds to Δy radians of angular separation in the center-of-mass frame—even when the lab-frame angle between them is milliradians.

Part (e): Why Rapidity Is Essential for Hadronic Collisions

What is this question really asking?

In e^+e^- collisions, using θ works fine. But in collisions involving hadrons (pp, ep, Au+Au, ...), rapidity becomes essential. Why?

Key hint: Inside a proton, each quark and gluon carries a **random fraction** x of the proton's momentum, and that fraction is different for every collision.

The e^+e^- baseline: life is easy

In e^+e^-, the electron and positron are elementary—each brings the full beam energy. Every event has the same center-of-mass frame, which (for a symmetric collider) is identical to the lab frame. The scattering angle θ measured in the lab *is* θ_{CM}. No ambiguity. Rapidity is nice but not essential.

The hadron collision problem: the CM frame is random

In a proton-proton collision, the *protons* don't actually collide—a quark from one proton hits a gluon (or quark) from the other. These partons carry momentum fractions x_1 and x_2, pulled randomly from the parton distribution functions (PDFs).

If proton 1 has beam momentum P, the colliding parton has momentum x_1P. From the other side, x_2P. The partonic center-of-mass frame is boosted relative to the lab by:

$$y_{\text{boost}} = \frac{1}{2}\ln\left(\frac{x_1}{x_2}\right)$$

Since x_1 and x_2 are *different in every event*, the boost y_{boost} is a **random number that changes from collision to collision**. Sometimes the CM is near the lab frame ($x_1 \approx x_2$); sometimes it's strongly boosted ($x_1 \gg x_2$). And you usually don't know which.

Why θ fails

If you measure the scattering angle θ in the lab, you're seeing a mixture of two effects tangled together:

1. The actual partonic scattering angle (the physics you want).
2. A random Lorentz boost from $x_1 \neq x_2$ (noise you don't want).

Two events with identical underlying physics will give *different* lab angles θ because of different random boosts. Averaging over many events smears the θ distribution in a complicated way that's very hard to undo.

Why rapidity works beautifully

Because $y' = y + y_{\text{boost}}$, the random event-by-event boost only *shifts* the rapidity of every particle by a constant. This means:

- Δy **between particles is invariant.** The opening angle of a jet, the separation between two leptons from a Z decay, the gap between two jets—none of these change, no matter what x_1/x_2 happened to be. You never need to reconstruct the parton kinematics.
- dN/dy **keeps its shape.** Each event's distribution is just shifted. Averaging over many events (i.e., over the PDFs) smooths the shifts into a broad, symmetric plateau—exactly what we observe in data.
- **The invariant cross section** $d^3\sigma/(dy\, d^2p_T)$ **is Lorentz invariant.** So the natural coordinates for presenting results are (y, p_T), not (θ, p).

In short: rapidity absorbs the unknowable, random parton boosts into a harmless overall shift, letting you do physics as if you were in the CM frame without ever needing to know where the CM frame actually was.

Answer

In hadronic collisions, the hard scattering happens between partons carrying random momentum fractions x_1 and x_2, so the parton–parton CM frame is randomly boosted by $y_{\text{boost}} = \frac{1}{2}\ln(x_1/x_2)$—a number that is different for every collision event and usually unknown.

Rapidity handles this gracefully: the random boost is just an additive shift ($y' = y + y_{\text{boost}}$), so Δy between particles is completely invariant and dN/dy distributions keep the same shape regardless. The angle θ, by contrast, gets scrambled differently by every event's random boost in a complicated, particle-dependent way that cannot be cleanly undone.

This is why rapidity is *essential* for hadron collisions but merely convenient for e^+e^-: in e^+e^-, the CM frame is known and fixed ($x_1 = x_2 = 1$), so there is no random boost to worry about.

Cheat Sheet

Formula Comparison

	Class (Collider) Def.	**Textbook Def.**
Rapidity y	$\operatorname{arctanh}(\beta\cos\theta)$	$\operatorname{arctanh}(\beta)$
Gives	$E = m_T\cosh y, \quad p_z = m_T\sinh y$	$E = m\cosh y, \quad p = m\sinh y$
They agree	Only when $p_T = 0$ (particle along beam)	

y vs. η vs. θ

Property	$\boldsymbol{y}$	$\boldsymbol{\eta}$	$\boldsymbol{\theta}$
Definition	$\operatorname{arctanh}(p_z/E)$	$\operatorname{arctanh}(p_z/\lvert\vec{p}\rvert)$	polar angle
Needs mass/energy?	Yes	**No**	**No**
Δ is Lorentz invariant	**Yes** (exact)	$\approx$ yes	**No**
$dN/d(\cdot)$ is flat	**Yes**	**Yes**	**No**
Units $\approx$ radians (COM)	**Yes**	**Yes**	trivially
Beam value is finite	**Yes**	**No** ($\to\infty$)	Yes ($=0$)
Handles random PDF boosts	**Yes**	$\approx$ yes	**No**

Key Facts to Memorize

Boost law	$y' = y + y_F$ (additive, like Galilean velocity!)
$y = \eta$ when	$m = 0$ (exact), or $E \gg m$ (approximate)
Radians connection	$\lvert d\eta \rvert = \lvert d\theta \rvert$ at $\theta = 90°$; extends everywhere by boost inv.
Partonic boost	$y_{\text{boost}} = \frac{1}{2}\ln(x_1/x_2)$ (random, event-by-event)
Invariant cross section	$E\,\frac{d^3\sigma}{d^3p} = \frac{d^3\sigma}{dy\,d^2p_T}$ (natural hadronic variables)